

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING (FTE)
Chandubhai S. Patel Institute of Technology (CSPIT) &
Devang Patel Institute of Advance Technology and Research (DEPSTAR)

ACADEMIC YEAR: 2024-25

Practical List

Subject: Cryptography and Network Security (IT348) (6th Sem)

Exp. No.	Name of Experiment	Hours
1.	A security training institute is setting up a lab for ethical hacking workshops. The team must decide whether to use Kali Linux or Parrot Security OS, prioritizing ease of installation, hardware requirements, and post-installation configuration for beginner and intermediate students. Evaluate and install the most suitable penetration testing operating system for a security team by exploring the installation processes and user-friendliness of Kali Linux and Parrot Security OS. References: 1. Kali OS: https://www.kali.org/ 2. Kali OS Installation: https://www.youtube.com/watch?v=jk2KGdJU2OI 3. Parrot Security OS: https://parrotsec.org/ 4. Parrot Security OS Installation: https://www.youtube.com/watch?v=4qvFp99rfXw	02
2.	The transmission of information needs to be secure over the communication channel and the data has to be confidential. To do so, steganography is the technique of concealing/hiding a secret file, message, audio, or video in another file format. Study and implement the practical approach for Steganography using the following tools: Steghide, StegoSuite & Xiao Steganography. Reference: https://www.youtube.com/watch?v=xepNoHgNj0w	02
3.	Footprinting is the process of accumulating data regarding a specific network environment, usually to find ways to intrude into it. It can reveal system vulnerabilities and improve the ease with which they can be exploited. Footprinting is also known as reconnaissance. Study a practical approach to implementing Footprinting: Gathering Target Information using the OSINT Framework. Reference: https://osintframework.com/	02
4.	Port scanning is a method for determining open ports and services available on a network or a host. It involves connecting with TCP and UDP ports on the system once you find the IP addresses of a target network or host using the Footprinting technique. You have to map the network of this targeted organization. Nmap (Network Mapper) is a powerful, flexible, open-source, easy-to-use port scanning tool available for both Linux and Windows-based operating	02

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	systems. Study practical approaches to implementing scanning and enumeration techniques using Nmap. Reference: https://www.youtube.com/watch?v=fp1042XK4A8	
5.	RSA algorithm is a public key encryption technology. It is considered the most secure way of encryption to secure sensitive data, particularly when sent over an insecure network such as the Internet. The public and private key generation algorithm is the most complex part of the RSA algorithm. The strength of RSA is the difficulty of factoring large integers that are the product of two large prime numbers, which is considered infeasible due to the time it would take using even today's highly configured computers. Implement the RSA algorithm. Reference: https://www.youtube.com/watch?v=VF3AHG0T9ec	02
6.	A medium-sized enterprise is concerned about the security posture of its internal network after a recent breach. The IT security team uses OpenVAS to conduct a comprehensive vulnerability assessment of the network's servers and devices to identify exploitable vulnerabilities. Perform a vulnerability scan on a network using OpenVAS and analyse the results to identify and mitigate potential security risks. Reference: 1. OpenVAS - https://www.openvas.org/ 2. Complete Beginner OpenVAS Vulnerability Scanning Tutorial - https://www.youtube.com/watch?v=LGH2SetiKaY 3. Perform a Vulnerability Scan Using OpenVAS - https://www.youtube.com/watch?v=lReXcE9BT0g 4. Vulnerability Scanning with OpenVAS TryHackMe – https://www.youtube.com/watch?v=DOJ2NeVa7i8	04
7.	A digital forensics team is investigating a case involving encrypted files and documents critical to their investigation. The team must recover the passwords for various applications, including archived files and PDF documents, using tools like Passware Password Recovery Kit Forensic, Advanced Archive Password Recovery, and Advanced PDF Password Recovery. Recover application passwords using specialised tools to demonstrate password recovery techniques and evaluate the efficiency of each tool in real-world scenarios. References: 1. Passware Password Recovery Kit Forensic - https://www.passware.com/ 2. Advanced Archive Password Recovery https://www.elcomsoft.com/archpr.html 3. Advanced PDF Password Recovery - https://www.elcomsoft.com/apdfpr.html	02

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8.	An organisation has transferred sensitive financial documents between departments over a network and wants to ensure that the files remain unchanged during transit. The IT team is tasked with using HashCalc and MD5 Calculator to confirm the integrity of these files by comparing hash values generated before and after the transfer. Verify the integrity of critical data files by generating and comparing hash values using HashCalc and MD5 Calculator, ensuring no tampering or corruption has occurred during transmission or storage. Reference: https://hashcalc.en.softonic.com/	02
9.	A cybersecurity training program is preparing students to understand and apply cryptographic techniques in real-world scenarios. The students are tasked with using CrypTool to study various cryptographic algorithms and simulate encryption and decryption processes to gain practical insights into data security mechanisms. To explore and analyse cryptographic algorithms using CrypTool to understand their functionality, strengths, and weaknesses in securing sensitive information. Reference: https://www.cryptool.org/en/	04
10.	A mid-sized company plans to enhance network security by deploying a robust firewall solution. The IT security team is tasked with studying and testing various firewall software options to determine the most suitable one based on functionality, ease of use, and performance. To evaluate and compare different firewall software solutions for securing network traffic and preventing unauthorised access in a simulated enterprise environment. Reference: Windows Defender Firewall - https://www.youtube.com/watch?v=pP7_nFBNR-M	04
11.	Wireshark is an open-source tool for profiling network traffic and analysing packets. It is often referred to as a network analyser, network protocol analyser, or sniffer. Wireshark intercepts traffic and converts that binary traffic into a human-readable format. Network administrators, Network security engineers, QA engineers, Developers, and other people who troubleshoot network problems, examine security problems, verify network applications, debug protocol implementations, and learn network protocol internals, respectively, can use it. A practical approach to study Wireshark from network security concept. Reference: https://www.youtube.com/watch?v=yC0e0bSSleo	04